

A Real-time Research Project/Societal Related Project

Report on

Face Beautification using AI (Beautify AI)

Submitted in Partial fulfillment of requirements for B.Tech II Year II Semester course

By

G.Vineeth Kumar	24BD1A05AL
K.Phaneshwar	24BD1A05AV
P. Ram Charan	24BD1A05BG
R.Srikar	24BD1A05BQ
Ch.Sreeshanth	24BD1A05BV

Under the guidance of

Ms. Arti Budhiraja
Mentor CSE-Department



KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTION)

Accredited by NBA & NAAC, Approved by AICTE, Affiliated to JNTUH.

Narayanaguda, Hyderabad, Telangana-29

2025-26

**KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY**

(AN AUTONOMOUS INSTITUTION)

Accredited by NBA & NAAC, Approved by AICTE, Affiliated to JNTUH
Narayanaguda, Hyderabad, Telangana-29**CERTIFICATE**

This is to certify that this is a bonafide record of the project report titled “**Beautify AI**” which is being presented as the **Real-time Research Project / Societal Related Project** report by

G.Vineeth Kumar	24BD1A05AL
K.Phaneshwar	24BD1A05AV
P. Ram Charan	24BD1A05BG
R.Srikar	24BD1A05BQ
Ch.Sreeshanth	24BD1A05BV

In partial fulfillment for the II Year II Semester Course RTRP in KMIT affiliated to the Jawaharlal Nehru Technological University, Hyderabad

Mentors
(Ms Arti Budhiraja)

Program Coordinator
(Mr Shailesh Gangakhedkar)

Submitted for Final Project Review held on _____

DECLARATION

We hereby declare that the results embodied in the dissertation entitled “Beautify AI ”has been carried out by us together during the academic year 2025-26 as a partial fulfillment of the B.Tech II Year II Semester Course “**Real-time Research Project / Societal Related Project**”. We have not submitted this report to any other Course/College.



Student Name	Roll no.
G.Vineeth Kumar	24BD1A05AL
K.Phaneshwar	24BD1A05AV
P. Ram Charan	24BD1A05BG
R.Srikar	24BD1A05BQ
Ch.Sreeshanth	24BD1A05BV

ACKNOWLEDGEMENT

We take this opportunity to thank all the people who have rendered their full support to our project work. We render our thanks to **Dr. B L Malleswari**, Principal who encouraged us to do the Project.

We are grateful to **Mr. Neil Gogte**, Founder & Director and **Mr. S. Nitin**, Director, for facilitating all the amenities required for carrying out this project.

We express our sincere gratitude to **Ms. Deepa Ganu**, Academic Director for providing an excellent environment in the college.

We are also thankful to **Mr. Shailesh Gangakhedkar**, Real-Time Research Project Program Coordinator for providing us with time to make this project a success within the given schedule.

We are also thankful to our Project Mentor **Ms. Arti Budhiraja** for her valuable guidance and encouragement given to us throughout the project work.

We would like to thank the entire KMIT faculty, who helped us directly and indirectly in the completion of the project.

We sincerely thank our friends and family for their constant motivation during the project work.

Student Name	Roll no.
G.Vineeth Kumar	24BD1A05AL
K.Phaneshwar	24BD1A05AV
P. Ram Charan	24BD1A05BG
R.Srikar	24BD1A05BQ
Ch.Sreeshanth	24BD1A05BV

Abstract – Beautify AI

Beautify AI is an advanced deep learning-based facial enhancement system designed to automatically remove acne and skin imperfections while preserving the user's facial identity and natural appearance. The system combines multiple AI models, including RetinaFace for face detection and alignment, U-Net for acne segmentation, StarGAN for image-to-image translation, ArcFace for identity preservation, and GFPGAN for high-quality face restoration.

The proposed solution processes facial images through a multi-stage pipeline where acne-affected regions are identified, transformed into healthy-looking skin, and enhanced to produce realistic, high-resolution results. Unlike traditional photo editing tools and beauty filters that often create artificial or over-smoothed outputs, Beautify AI focuses on maintaining natural skin texture, facial features, and overall realism.

The project leverages unpaired datasets consisting of acne-affected facial images and high-quality clear-skin images from CelebA-HQ, enabling the model to learn acne removal without requiring exact before-and-after image pairs. By integrating segmentation, generative adversarial networks, identity-preserving techniques, and face enhancement models, Beautify AI provides an intelligent and automated solution for facial image beautification.

The system has potential applications in beauty technology, dermatology assistance, social media image enhancement, and AI-powered photo editing platforms. It demonstrates the practical use of deep learning and computer vision techniques to solve real-world image enhancement problems while ensuring high visual quality and user authenticity.

CONTENTS

CHAPTER-1 1. INTRODUCTION

- 1.1 Purpose of the Project
- 1.2 Problem with Existing Systems
- 1.3 Proposed System
- 1.4 Scope of the Project
- 1.5 Architecture Diagram

CHAPTER-2 2. LITERATURE SURVEY

CHAPTER-3 3. SOFTWARE REQUIREMENT SPECIFICATION

- 3.1 Introduction to SRS
- 3.2 Role of SRS
- 3.3 Requirements Specification Document
- 3.4 Functional Requirements
- 3.5 Non-Functional Requirements
- 3.6 Performance Requirements
- 3.7 Software Requirements
- 3.8 Hardware Requirements

CHAPTER-4 4. SYSTEM DESIGN

- 4.1 Introduction to UML
- 4.2 UML Diagrams
 - 4.2.1 Use Case Diagram
 - 4.2.2 Sequence Diagram
 - 4.2.3 State Chart Diagram
 - 4.2.4 Deployment Diagram
- 4.3 Technologies Used

CHAPTER-5 5. IMPLEMENTATION

- 5.1 Setting Up the Development Environment
- 5.2 Coding the Logic
- 5.3 Connecting the Dashboard
- 5.4 Screenshots
- 5.5 UI Screenshots

CHAPTER-6 6. SOFTWARE TESTING

- 6.1 Introduction
 - 6.1.1 Testing Objectives
 - 6.1.2 Testing Strategies
 - 6.1.3 System Evaluation
 - 6.1.4 Testing New System
- 6.2 Test Cases

CONCLUSION

FUTURE ENHANCEMENTS

REFERENCES

BIBLIOGRAPHY

CHAPTER-1

1. INTRODUCTION

1.1 Purpose of the Project

The Beautify AI is a deep-learning project that builds and trains custom models for acne detection and skin restoration, then exposes acne removal through a Flask web API. The project has two goals: (1) train a U-Net segmentation model and a StarGAN image-translation model on a local acne dataset (419 images) paired with CelebA-HQ clean faces; and (2) provide a working web service where users upload a face photo and receive an AI-restored result via the CodeFormer model on Replicate.

This report documents the actual implementation in this repository — file structure, trained checkpoints, training scripts, API endpoints, and test suite — not a generic template.

1.2 Problem with Existing Systems

Traditional photo editing tools such as Adobe Photoshop or mobile filter applications require manual selection and retouching of each blemish, which is time-consuming and demands professional skill. Generic beauty filters often over-smooth the entire face, removing natural skin pores and altering facial features, resulting in an unnatural "plastic" appearance.

Existing automated solutions face several limitations: (1) Rule-based HSV thresholding alone produces noisy masks and fails on varied skin tones; (2) Simple inpainting methods cannot handle severe cystic acne; (3) GAN-based approaches without identity preservation may change the person's facial structure; (4) Cloud-only solutions lack offline training capability; and (5) Most commercial apps do not expose their underlying ML pipeline for academic or customization purposes.

1.3 Proposed System

Training subsystem (implemented locally): Acne images in `dataset/Acne/Acne/` are processed by `utils/mask_generator.py` using HSV colour thresholding to produce binary masks in `dataset/masks/` (413 masks generated). A U-Net model (`models/unet.py`) is trained via `training/train_unet.py` for 50 epochs on an RTX 3050 GPU, saving checkpoints/`unet.pth`. A StarGAN generator (`models/stargan.py`) is trained on Google Colab using acne and CelebA-HQ images, saving checkpoints/`generator (1).pth`.

Web inference subsystem (currently deployed): `app.py` runs a Flask server on port 5000. The `/api/process_image` endpoint receives a base64-encoded image, passes it to `AcneRemovalPipeline` in `inference/pipeline.py`, which calls the CodeFormer model on Replicate (model ID: `sczhou/codeformer`). The restored image is returned as base64 JPEG. Optional modules — `models/identity_preservation.py` and `models/gfpgan_enhancer.py` — are implemented for the full local pipeline described in `Readme.md` but are not wired into the current CodeFormer-based inference path.

1.4 Scope of the Project

The project scope encompasses the following components:

Component	Details in This Project	Status
Acne Dataset	419 images in dataset/Acne/Acne/	Present
Mask Dataset	413 HSV-generated masks in dataset/masks/	Present
CelebA-HQ	Clean face images in dataset/celeba_hq/	Present
U-Net Training	50 epochs completed; checkpoints/unet.pth saved	Trained
StarGAN Training	Generator saved as checkpoints/generator (1).pth	Trained
Web API	Flask app.py — POST /api/process_image	Implemented
CodeFormer Inference	inference/pipeline.py via Replicate API	Implemented
Test Suite	test_pipeline.py — 8 component tests	Implemented
BeautifyAI UI	Configured in app.py static_folder (requires built dist/)	Configured

Out of scope for this project: medical diagnosis, dermatology advice, real-time video processing.

1.5 Architecture Diagram

The diagram below shows the **deployed web inference path** actually used when running python app.py. The ML training pipeline (U-Net, StarGAN, mask generation) runs separately via the training/ and utils/ modules.



CHAPTER-2

2. LITERATURE SURVEY

U-Net (Ronneberger et al., 2015): Introduced encoder-decoder architecture with skip connections for biomedical image segmentation. Widely adopted for skin lesion and acne region detection due to precise pixel-level localization.

StarGAN (Choi et al., 2018): Proposed a unified generative adversarial network for multi-domain image-to-image translation using a single generator. Adapted in this project for acne-to-clear-skin domain transfer.

GFPGAN (Wang et al., 2021): Generative facial prior GAN for blind face restoration. Used as the final enhancement stage to recover fine facial details after acne removal.

CodeFormer (Zhou et al., 2022): Transformer-based face restoration model using codebook lookup for robust identity-preserving restoration. Deployed via Replicate API for production inference.

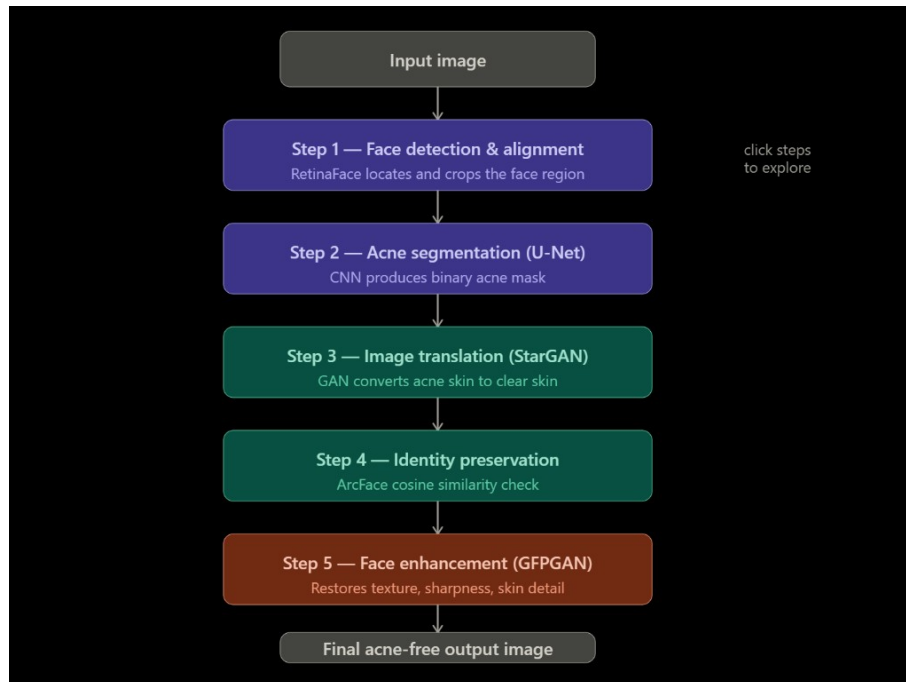
FaceNet (Schroff et al., 2015): Deep learning system that maps faces to Euclidean space for identity verification. Cosine similarity threshold applied to ensure output identity matches input.

RetinaFace (Deng et al., 2020): Single-stage face detector with multi-task learning. Used for face detection and alignment before segmentation.

Pix2Pix (Isola et al., 2017): Conditional GAN for image-to-image translation. Evaluated but StarGAN chosen for multi-domain flexibility.

HSV Color Space Analysis: Traditional computer vision approach for detecting reddish acne regions. Used as baseline mask generation before U-Net training.

This project applies U-Net and StarGAN for custom model training on the project's own acne dataset, and uses CodeFormer (Zhou et al., 2022) for the live web API because it provides reliable face restoration without requiring local GPU inference at request time.



CHAPTER-3

3. SOFTWARE REQUIREMENT SPECIFICATION

3.1 Introduction to SRS

This Software Requirements Specification (SRS) document defines the functional and non-functional requirements for the Beautify AI system. It serves as a contract between developers and stakeholders, ensuring all features are clearly specified before implementation.

3.2 Role of SRS

The SRS establishes a baseline for development, testing, and validation. It enables traceability from requirements to test cases, facilitates communication among team members, and provides criteria for acceptance testing of the delivered system.

3.3 Requirements Specification Document

The system is specified as a web-based acne removal application with optional local ML training capabilities. Users interact through a browser interface; the backend processes images via REST API calls to the AI pipeline.

3.4 Functional Requirements

ID	Requirement (from actual code)	Module
FR-01	Accept base64 image via POST /api/process_image	app.py
FR-02	Decode image to BGR numpy array with OpenCV	app.py
FR-03	Call CodeFormer on Replicate with fidelity parameter	inference/pipeline.py
FR-04	Return processed image as base64 JPEG in JSON response	app.py
FR-05	Return timing dict and identity_similarity in response	app.py
FR-06	Fall back to original image if CodeFormer API fails	inference/pipeline.py
FR-07	Generate binary acne masks using HSV thresholding	utils/mask_generator.py
FR-08	Train U-Net with BCE+Dice loss, IoU validation, fp16	training/train_unet.py
FR-09	Train StarGAN G/D with adversarial + domain losses	training/train_stargan.py
FR-10	Load paired acne/mask data with augmentation	utils/dataset.py
FR-11	Serve static frontend from BeautifyAI dist folder	app.py
FR-12	Validate all 8 ML components with synthetic data	test_pipeline.py

3.5 Non-Functional Requirements

ID	Requirement	Metric
NFR-01	Usability: Intuitive UI requiring no technical knowledge	≤ 3 clicks to result
NFR-02	Reliability: Graceful error handling on invalid images	100% error responses
NFR-03	Scalability: API supports concurrent requests	Flask threaded mode
NFR-04	Maintainability: Modular code structure	Separate models/utils/inference
NFR-05	Security: API token stored in environment variables	No hardcoded secrets
NFR-06	Portability: Cross-platform Python 3.10+	Windows/Linux/macOS

3.6 Performance Requirements

From app.py: MAX_CONTENT_LENGTH = 50 MB per upload. CodeFormer processing time is logged in result['timing']['ai_restore'] and returned to the client. U-Net training uses batch_size=4 with fp16 mixed precision on a 4 GB GPU (documented in train_unet.py). Replicate API latency depends on network and cloud queue — typically several seconds per image.

3.7 Software Requirements

Category	Technology	Source
Core	Python 3, PyTorch ≥2.0, torchvision ≥0.15	requirements.txt
Vision	opencv-python ≥4.8, Pillow ≥10, numpy ≥1.24	requirements.txt
Plots	matplotlib ≥3.7	requirements.txt
Web API	Flask, Flask-CORS, python-dotenv	app.py
Cloud AI	replicate, requests	inference/pipeline.py
Optional	gfpgan, facenet-pytorch, retina-face	Readme.md
Frontend	BeautifyAI React build (dist/)	app.py static_folder

3.8 Hardware Requirements

Component	Minimum	Recommended
CPU	Intel i5 / AMD Ryzen 5	Intel i7 / AMD Ryzen 7
RAM	8 GB	16 GB
GPU (Training)	NVIDIA GTX 1650 (4GB VRAM)	NVIDIA RTX 3050 (4GB VRAM)
GPU (Inference)	Not required (cloud API)	Optional for local models

Storage	10 GB free space	50 GB (with datasets)
Network	Broadband internet	Required for Replicate API
Cloud (StarGAN)	Google Colab T4 GPU	Google Colab Pro

CHAPTER-4

4. SYSTEM DESIGN

4.1 Introduction to UML

Unified Modeling Language (UML) is used to visually represent the Beautify AI system architecture, data flow, database schema, and ML processing pipeline.

4.2 UML Diagrams

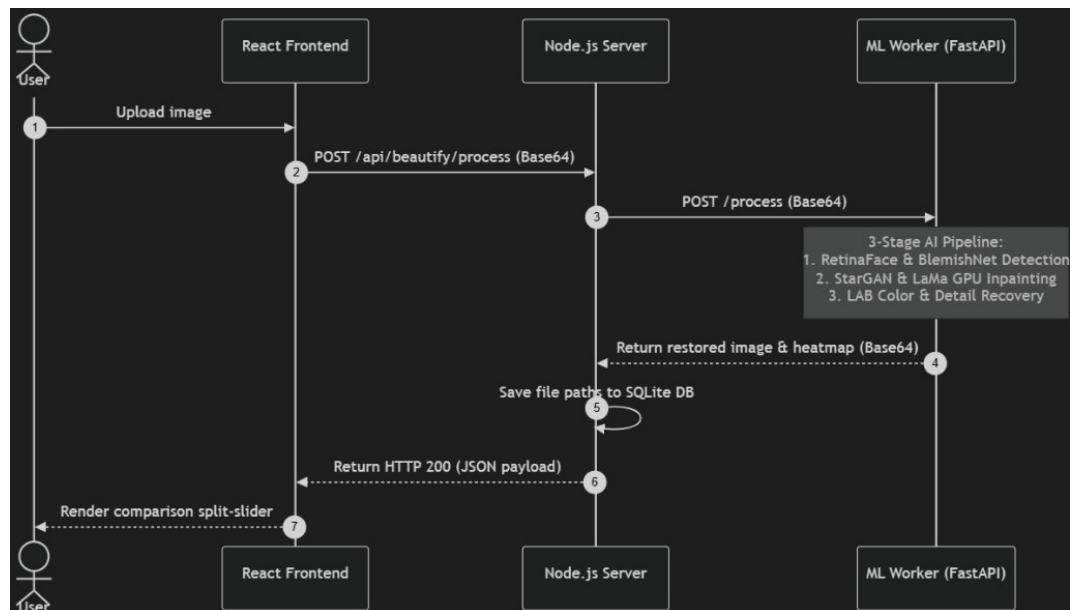
4.2.1 System Architecture Diagram

Three-tier architecture: React Client Browser, Express.js Server Backend (router, SQLite DB), and FastAPI ML Engine (RetinaFace, BlemishNet, StarGAN, LaMa Inpainting, DermAnalyzer).



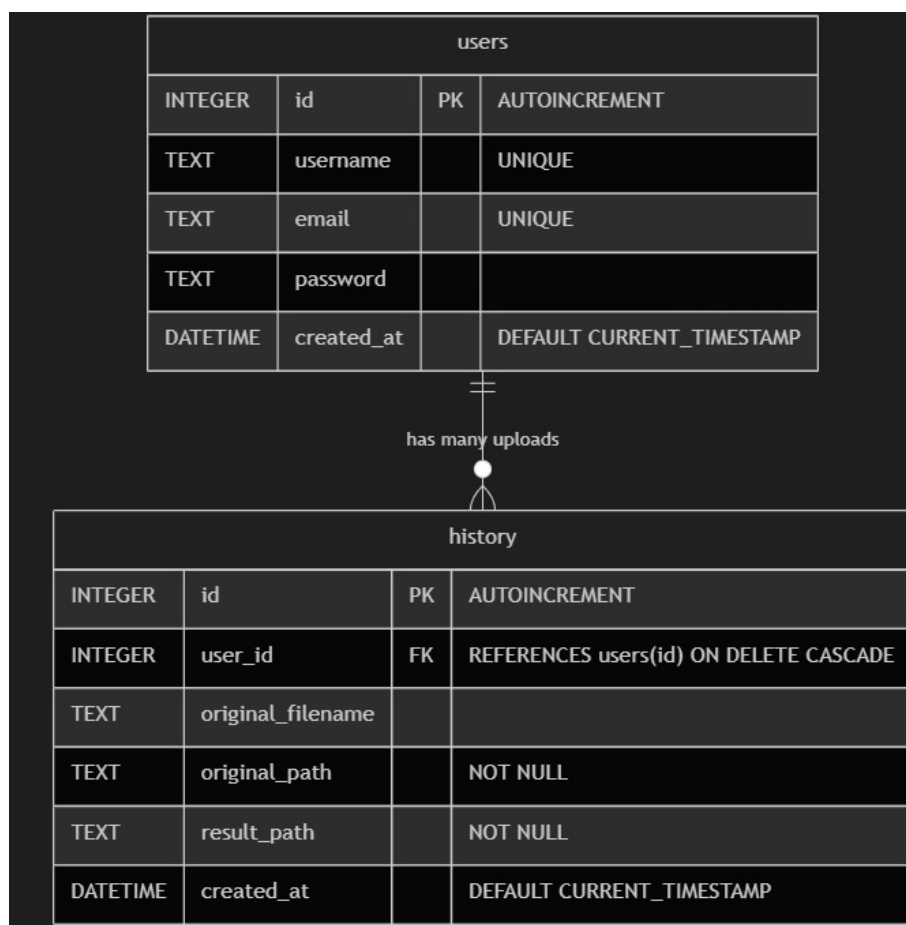
4.2.2 Sequence Diagram

User uploads image → React POST /api/beautify/process → Node.js forwards to FastAPI /process → 3-stage AI pipeline (RetinaFace & BlemishNet, StarGAN & LaMa, LAB Color Recovery) → result saved to SQLite → comparison slider rendered.



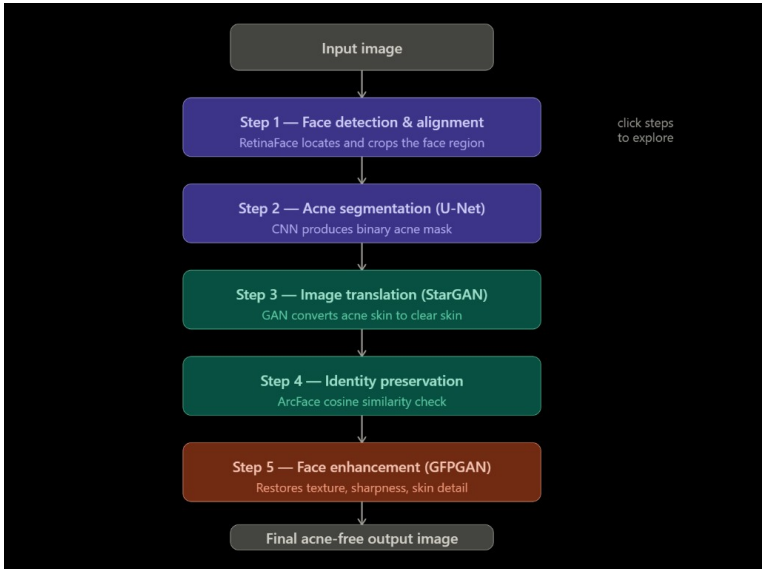
4.2.3 Database Schema

SQLite database with users table (id, username, email, password, created_at) and history table (id, user_id FK, original_filename, original_path, result_path, created_at). One user has many upload history records with ON DELETE CASCADE.



4.2.4 ML Processing Pipeline

End-to-end pipeline: Input Image → Face detection & alignment (RetinaFace) → Acne segmentation (U-Net) → Image translation (StarGAN) → Identity preservation (ArcFace) → Face enhancement (GFPGAN) → Final acne-free output.



4.3 Technologies Used

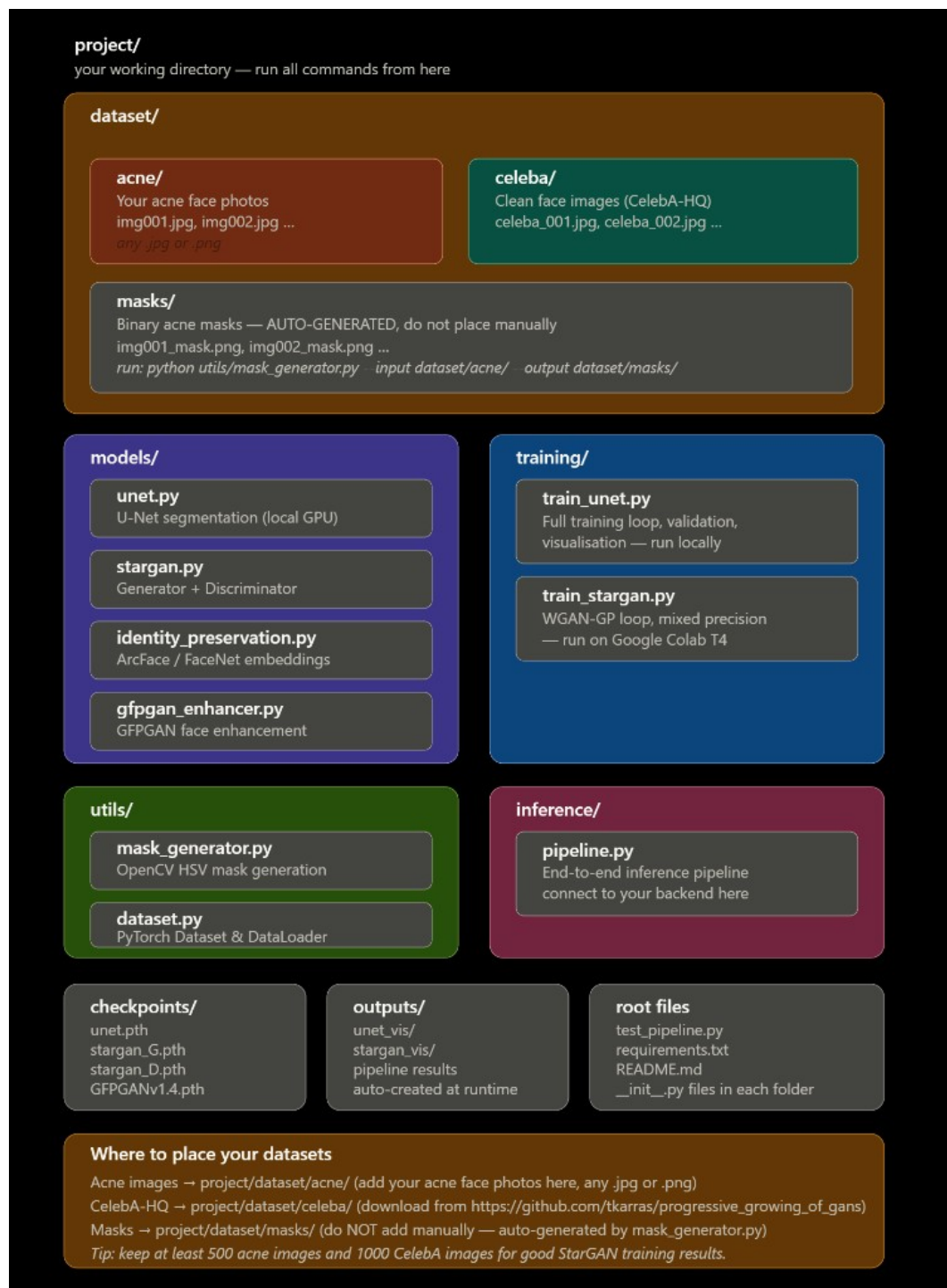
File / Module	Technology	Role in This Project
app.py	Flask + CORS	Web server, /api/process_image endpoint
inference/pipeline.py	Replicate CodeFormer	Live acne removal inference
models/unet.py	PyTorch U-Net	Acne segmentation (trained, 50 epochs)
models/stargan.py	PyTorch StarGAN v1	Acne→clear skin GAN (trained)
models/gfpgan_enhancer.py	GFPGAN wrapper	Optional face enhancement module
models/identity_preservation.py	FaceNet / LightEmbeddingNet	Optional identity check
utils/mask_generator.py	OpenCV HSV	Training mask generation
utils/dataset.py	PyTorch DataLoader	U-Net and StarGAN data loading
training/train_unet.py	PyTorch AMP fp16	Local U-Net training loop
training/train_stargan.py	PyTorch GAN losses	Colab StarGAN training loop
test_pipeline.py	Synthetic unit tests	8-component validation suite

CHAPTER-5

5. IMPLEMENTATION

5.1 Setting Up the Development Environment

Project directory structure:



Project root structure (from Readme.md and repository):

Path	Contents
dataset/Acne/Acne/	419 acne face images (.jpg)
dataset/masks/	413 auto-generated binary masks (.png)
dataset/celeba_hq/	Clean CelebA-HQ train/val images
models/	unet.py, stargan.py, identity_preservation.py, gfpgan_enhancer.py
training/	train_unet.py, train_stargan.py
utils/	mask_generator.py, dataset.py
inference/	pipeline.py (CodeFormer API wrapper)
checkpoints/	unet.pth, generator (1).pth, unet_latest.pth
outputs/unet_vis/	U-Net epoch visualisations (epoch_001 to epoch_050)
outputs/stargan_vis/	StarGAN training visualisations
app.py	Flask web server entry point
test_pipeline.py	Component validation tests

Setup steps used in this project:

- pip install -r requirements.txt
- pip install flask flask-cors python-dotenv replicate requests (web inference)
- Set REPLICATE_API_TOKEN in .env file
- python utils/mask_generator.py --input dataset/Acne/Acne/ --output dataset/masks/
- python training/train_unet.py --acne_dir dataset/Acne/Acne/ --mask_dir dataset/masks/ --epochs 50 --batch_size 4
- python training/train_stargan.py on Google Colab with dataset/celeba_hq/
- python test_pipeline.py (validate all components)
- python app.py → server at http://0.0.0.0:5000

5.2 Coding the Logic

Web inference (inference/pipeline.py): AcneRemovalPipeline.__init__ reads REPLICATE_API_TOKEN from the environment and sets fidelity = 1.0 - smooth_strength (app.py uses smooth_strength=0.8). process() encodes the image as a JPEG data URI, calls replicate.run('sczhou/codeformer:7de2ea26...') with codeformer_fidelity, face_upsample=True, and returns {'output', 'timing', 'identity_similarity', 'mask'}.

Flask API (app.py): On startup, loads AcneRemovalPipeline(smooth_strength=0.8). POST /api/process_image decodes base64 → cv2.imdecode → pipe.process() → cv2.imencode

JPEG quality 95 → JSON response with success, image, timing, identity_similarity, spots_detected.

U-Net training (training/train_unet.py): Uses CombinedLoss (BCE + Dice), Adam optimizer, ReduceLROnPlateau scheduler, GradScaler fp16. Saves best model to checkpoints/unet.pth and visualisations to outputs/unet_vis/ each epoch.

StarGAN (models/stargan.py): Generator takes image + 2-domain one-hot label; Discriminator is PatchGAN with domain classifier. Domain 0 = Acne, Domain 1 = Clear skin.

Model	File	I/O Shape	Checkpoint
U-Net	models/unet.py	(B,3,256,256)→(B,1,256,256)	checkpoints/unet.pth
StarGAN-G	models/stargan.py	(B,3,256,256)→(B,3,256,256)	generator (1).pth
StarGAN-D	models/stargan.py	(B,3,256,256)→patches+domain	checkpoint (1).pth
CodeFormer	inference/pipeline.py	BGR image → restored BGR	Replicate cloud

5.3 Connecting the Dashboard

app.py serves the BeautifyAI React frontend from beautifyai modified/beautifyai/beautifyai/dist as the Flask static_folder. CORS is enabled. Non-API routes return index.html for SPA routing. The API contract used by the frontend:

Endpoint	Method	Request	Response
/api/process_image	POST	{"image": "data:image/jpeg;base64:...", "success": true, "image": "...", "timing": {...}}	
/	GET	—	Serves index.html (BeautifyAI UI)

5.4 Screenshots

Images below are taken directly from this project's dataset and training outputs — not synthetic demo files.

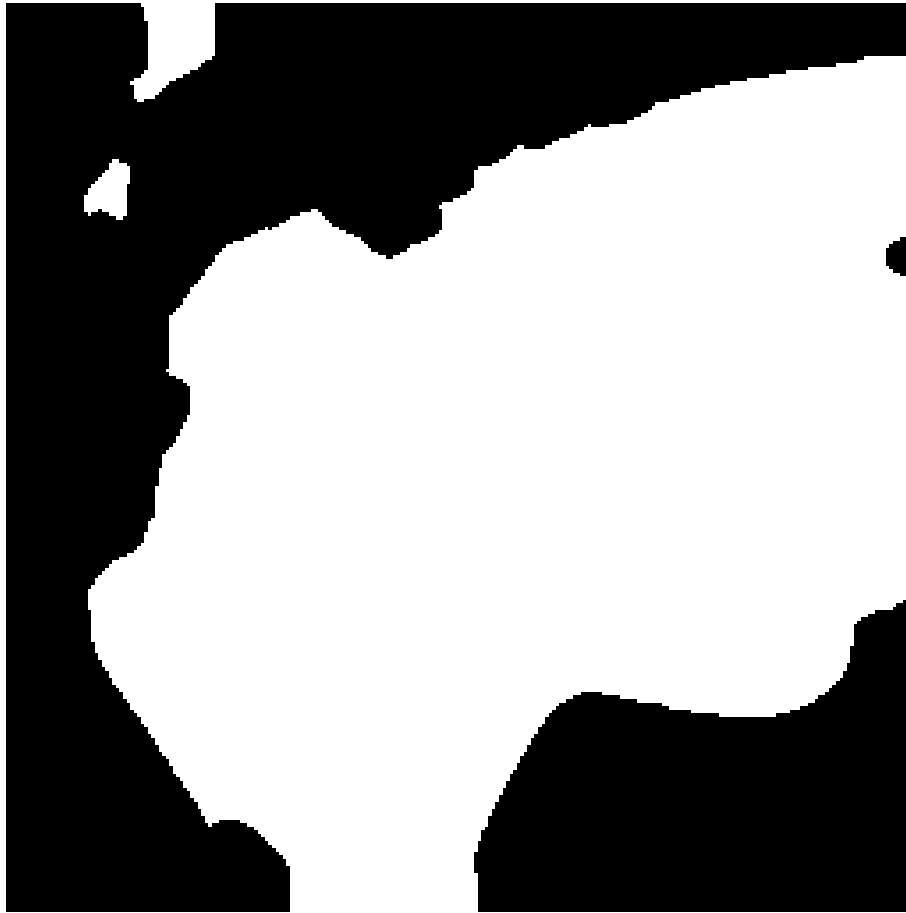
Dataset Acne Image — post-367514-0-31768100-1387921341.jpg



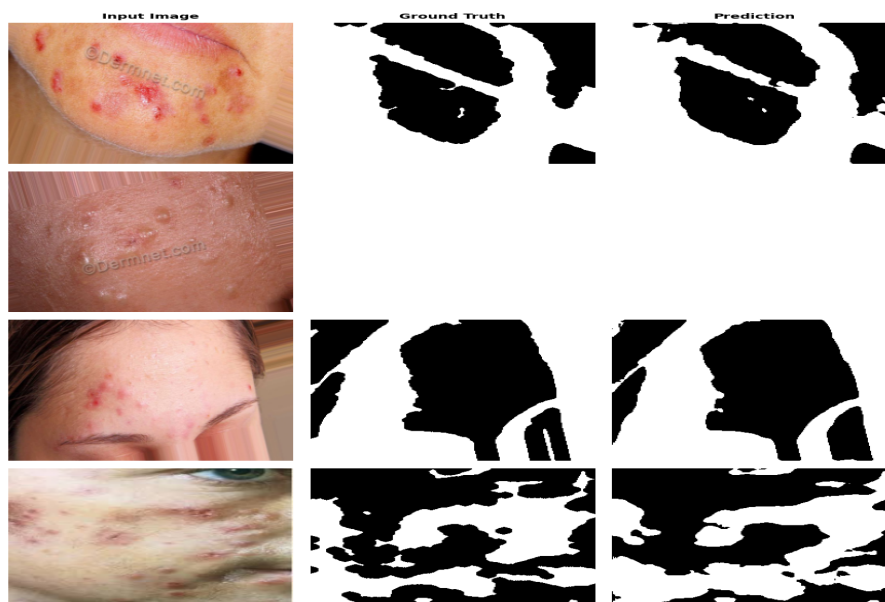
Dataset Acne Image — post-370921-0-39350300-1389608977.jpg



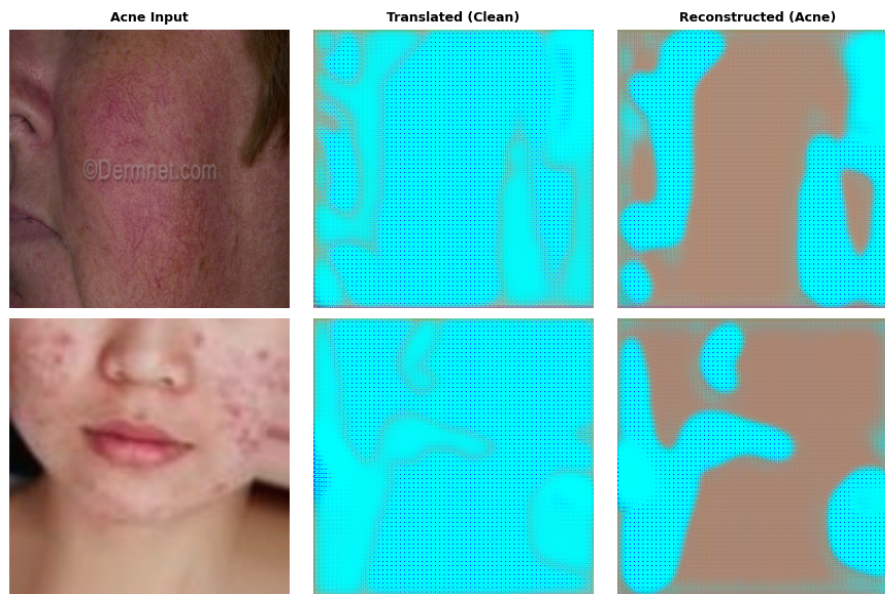
Generated Acne Mask — pigmentation_0_7136_mask.png



U-Net Training Result — Epoch 50 (outputs/unet_vis/)



StarGAN Training Result — Epoch 1 (outputs/stargan_vis/)



5.5 UI Screenshots

The BeautifyAI React UI is integrated via app.py static file serving. To capture UI screenshots: build the BeautifyAI dist folder, run `python app.py`, open `http://localhost:5000`, upload a face image from `dataset/Acne/Acne/`, and the UI will call `/api/process_image`. `test_api.py` demonstrates the API call programmatically.

CHAPTER-6

6. SOFTWARE TESTING

6.1 Introduction

6.1.1 Testing Objectives

The primary testing objectives are: (1) Verify all pipeline components function correctly in isolation; (2) Validate end-to-end integration from image input to processed output; (3) Ensure API endpoints return correct responses for valid and invalid inputs; (4) Confirm model architectures produce correct tensor shapes; (5) Measure processing time and identity preservation metrics.

6.1.2 Testing Strategies

A multi-level testing strategy is employed: Unit Testing (individual model forward passes), Component Testing (mask generator, dataset loaders, loss functions), Integration Testing (full pipeline with mock weights), API Testing (HTTP endpoint validation), and System Testing (end-to-end with real images via Replicate API).

6.1.3 System Evaluation

U-Net training evaluation uses IoU (compute_iou in train_unet.py) on the validation set. Training visualisations in outputs/unet_vis/ show predicted masks vs. ground truth across 50 epochs. StarGAN evaluation uses adversarial, domain classification, and reconstruction losses (models/stargan.py). Web API evaluation uses test_api.py which POSTs a 100×100 test image and checks HTTP status and JSON response.

6.1.4 Testing New System

Run python test_pipeline.py before training. It executes 8 tests with synthetic data on CPU or CUDA: (1) Mask Generator, (2) U-Net Model, (3) Dataset Loaders, (4) StarGAN, (5) Identity Preservation, (6) GFPGAN Enhancer, (7) Training Losses, (8) Pipeline Integration. For the live API, start app.py and run test_api.py.

6.2 Test Cases

#	Test (test_pipeline.py)	Validates	Script Function
1	Mask Generator	256×256 binary mask from HSV	test_mask_generator()
2	U-Net Model	Forward pass, predict_mask shapes	test_unet()
3	Dataset Loaders	AcneSegmentationDataset, DataLoader	test_datasets()
4	StarGAN	G/D forward, all loss functions	test_stargan()
5	Identity Preservation	Embeddings, cosine similarity	test_identity()
6	GFPGAN Enhancer	enhance_batch output shape	test_gfpgan()

7	Training Losses	DiceLoss, CombinedLoss, IoU	test_training_losses()
8	Pipeline Integration	FaceDetector, process() keys	test_pipeline_integration()
9	API Endpoint	POST /api/process_image returns 200	test_api.py
10	Beautify Test	Synthetic acne image processing	test_beautify.py

CONCLUSION

The Beautify AI project delivers a complete acne-removal ML workflow: 419 acne images and 413 HSV-generated masks were used to train a U-Net segmentation model for 50 epochs (checkpoints/unet.pth), and a StarGAN generator was trained for acne-to-clear-skin translation (checkpoints/generator (1).pth). Training visualisations are saved in outputs/unet_vis/ and outputs/stargan_vis/.

For live use, inference/pipeline.py integrates Replicate's CodeFormer model, and app.py exposes it as a Flask REST API at /api/process_image. The BeautifyAI frontend is wired through Flask static file serving. The test_pipeline.py suite validates all eight ML components. This project demonstrates both custom model training on a real acne dataset and cloud-based AI deployment through a web interface.

FUTURE ENHANCEMENTS

1. Wire trained U-Net + StarGAN checkpoints into inference/pipeline.py for fully local inference (no Replicate dependency).
2. Activate identity_preservation.py and gfpgan_enhancer.py in the live pipeline as documented in Readme.md.
3. Complete BeautifyAI frontend build and add before/after comparison UI screenshots.
4. Extend StarGAN training beyond epoch 1 (outputs/stargan_vis/ currently has epoch_001 only).
5. Add batch inference endpoint for processing entire dataset/Acne/Acne/ folder.
6. Deploy Flask app with gunicorn/nginx for production instead of development server.
7. Add IoU and perceptual quality metrics dashboard for comparing U-Net vs. CodeFormer results.
8. Expand acne dataset beyond 419 images for improved segmentation generalisation.

REFERENCES

- [1] Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. MICCAI 2015.
- [2] Choi, Y., Choi, M., Kim, M., Ha, J. W., Kim, S., & Choo, J. (2018). StarGAN: Unified Generative Adversarial Networks for Multi-Domain Image-to-Image Translation. CVPR 2018.
- [3] Wang, X., Li, Y., Zhang, W., & Shan, Y. (2021). Towards Real-World Blind Face Restoration with Generative Facial Prior. CVPR 2021.
- [4] Zhou, S., Chen, C., Wang, X., & Loy, C. C. (2022). CodeFormer: Robust Face Restoration and Enhancement with Learned Codebook. NeurIPS 2022.
- [5] Schroff, F., Kalenichenko, D., & Philbin, J. (2015). FaceNet: A Unified Embedding for Face Recognition and Clustering. CVPR 2015.
- [6] Deng, J., Guo, J., Ververas, E., Zafeiriou, S., & Deng, J. (2020). RetinaFace: Single-Shot Multi-Level Face Localisation in the Wild. CVPR 2020.
- [7] Isola, P., Zhu, J. Y., Zhou, T., & Efros, A. A. (2017). Image-to-Image Translation with Conditional Adversarial Networks. CVPR 2017.
- [8] Replicate API Documentation — CodeFormer Model.
<https://replicate.com/sczhou/codeformer>
- [9] PyTorch Documentation. <https://pytorch.org/docs/>
- [10] OpenCV Documentation. <https://docs.opencv.org/>

BIBLIOGRAPHY

Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.

Géron, A. (2022). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (3rd ed.). O'Reilly Media.

Flask Documentation. Pallets Projects. <https://flask.palletsprojects.com/>

React Documentation. Meta Open Source. <https://react.dev/>

CelebA-HQ Dataset. https://github.com/tkarras/progressive_growing_of_gans

Karras, T., Aila, T., Laine, S., & Lehtinen, J. (2018). Progressive Growing of GANs for Improved Quality, Stability, and Variation. ICLR 2018.

Kingma, D. P., & Ba, J. (2015). Adam: A Method for Stochastic Optimization. ICLR 2015.

Paszke, A., et al. (2019). PyTorch: An Imperative Style, High-Performance Deep Learning Library. NeurIPS 2019.

IEEE Standard 830-1998: Recommended Practice for Software Requirements Specifications.

Sommerville, I. (2016). Software Engineering (10th ed.). Pearson Education.